Cairo University

Faculty of Computers and Artificial Intelligence

CS251

Introduction to Software Engineering

$penderman

Project Report

Version 1.1

Apr-2026

Project Repository

- Repo Link

Team members Profiles

Ahmed Shawareb

- GitHub
- Linkedin

Ahmed Salah

- GitHub
- Linkedin

Mohamed Ayser

- GitHub
- Linkedin

Project description

$penderman is a smart personal budgeting desktop application built to give users crystal-clear visibility into their cash flow. Users can effortlessly manage multiple wallets, track customized expenses and deposits, and seamlessly fund their financial milestones through a secure, reactive, and data-driven dashboard.

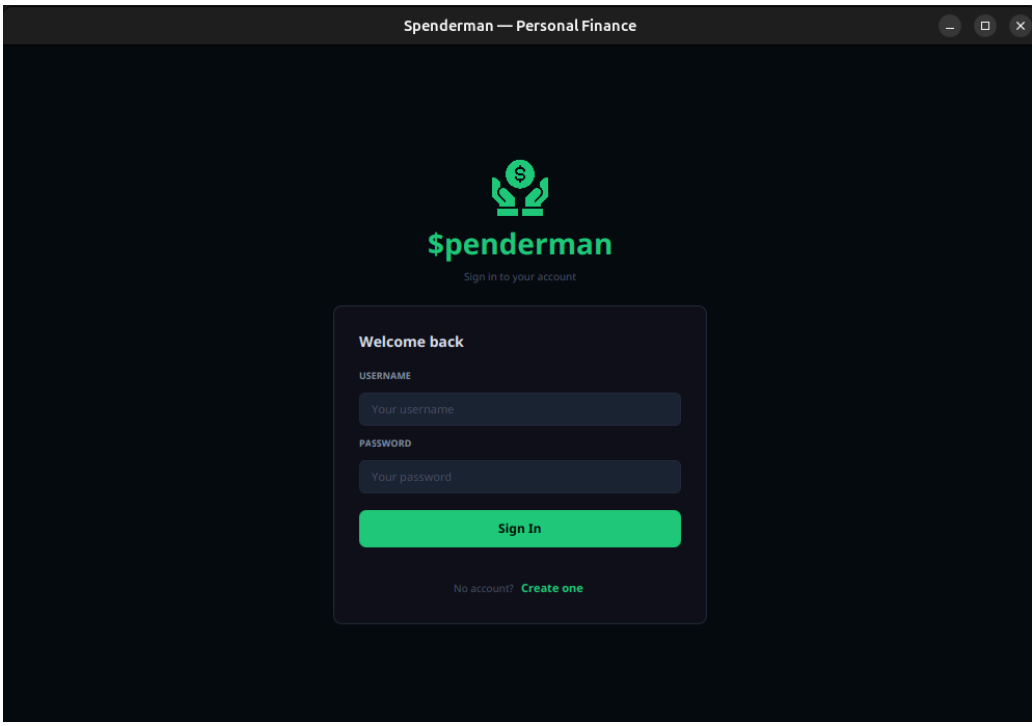
Ownership Report

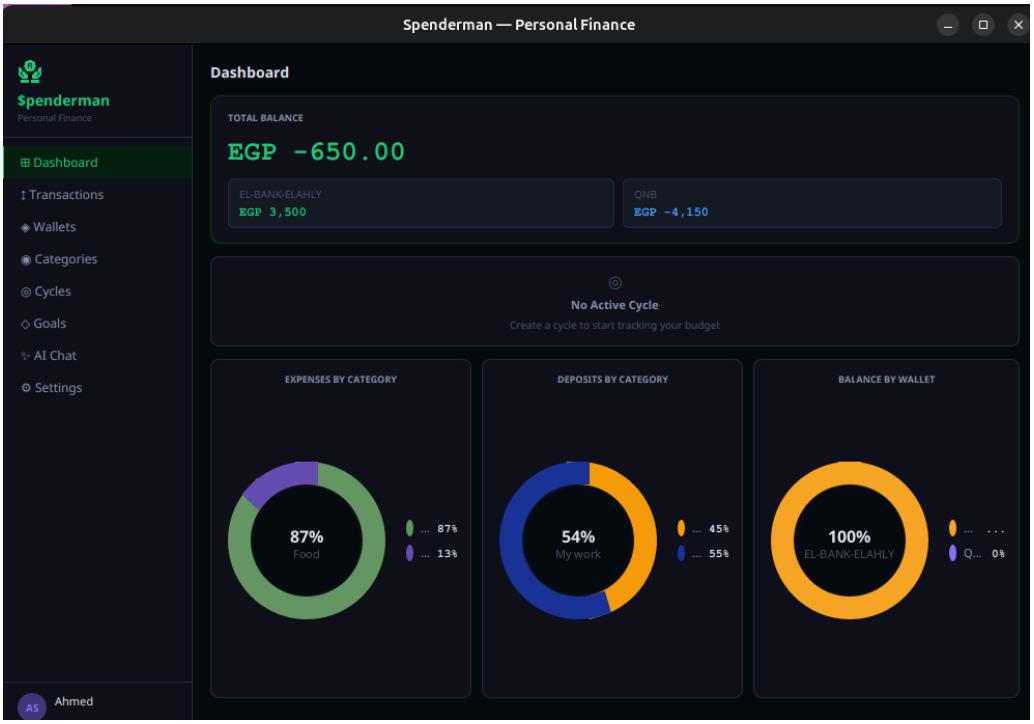
Item	ID	Owners
Ahmed Amr Abdelbadea Ali Shawareb	20240038	ER-Diagram , Schema Diagram , UML class Diagram , 3 Sequence diagram , Database hosting and creation , work in DAO layer , work in service layer
Ahmed Mohamed Salah El-Din	20240049	ER-Diagram , UML class Diagram , UI design , 3 sequence Diagram , GUI implementation , work in DAO layer , work in service layer
Mohamed Ayser Mohamed El-Hosiny	20240476	ER-Diagram , 2 State Diagrams , 3 Sequence Diagram , UML class Diagram , AI bot , work in DAO layer , work in service layer

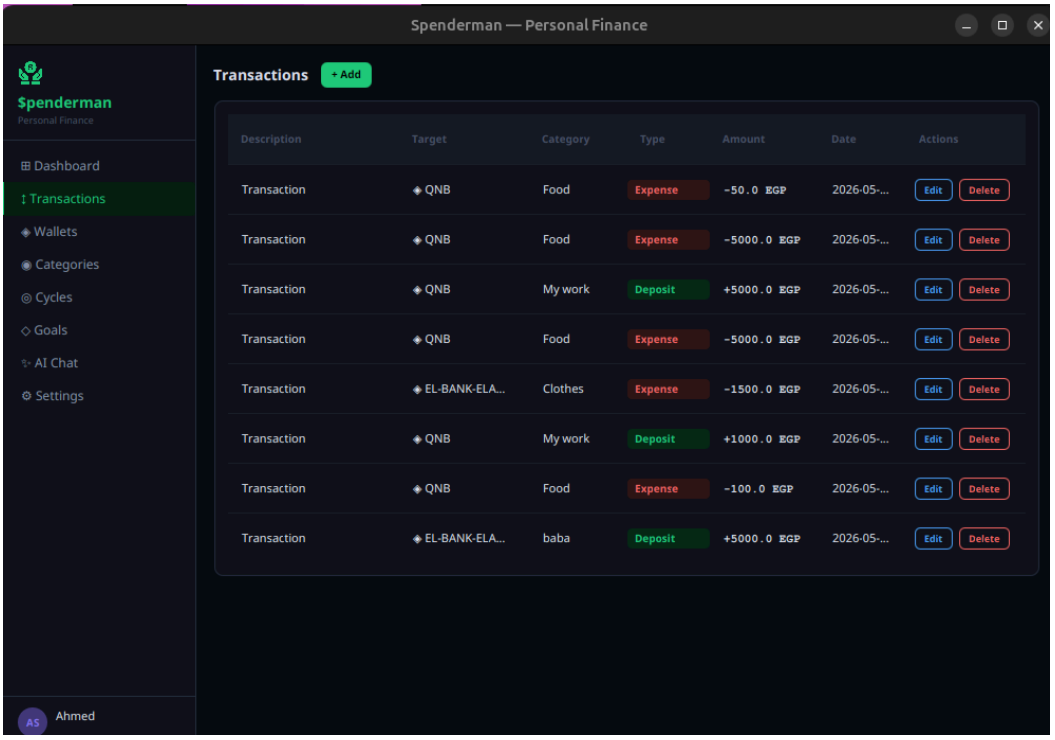
Bonus Features

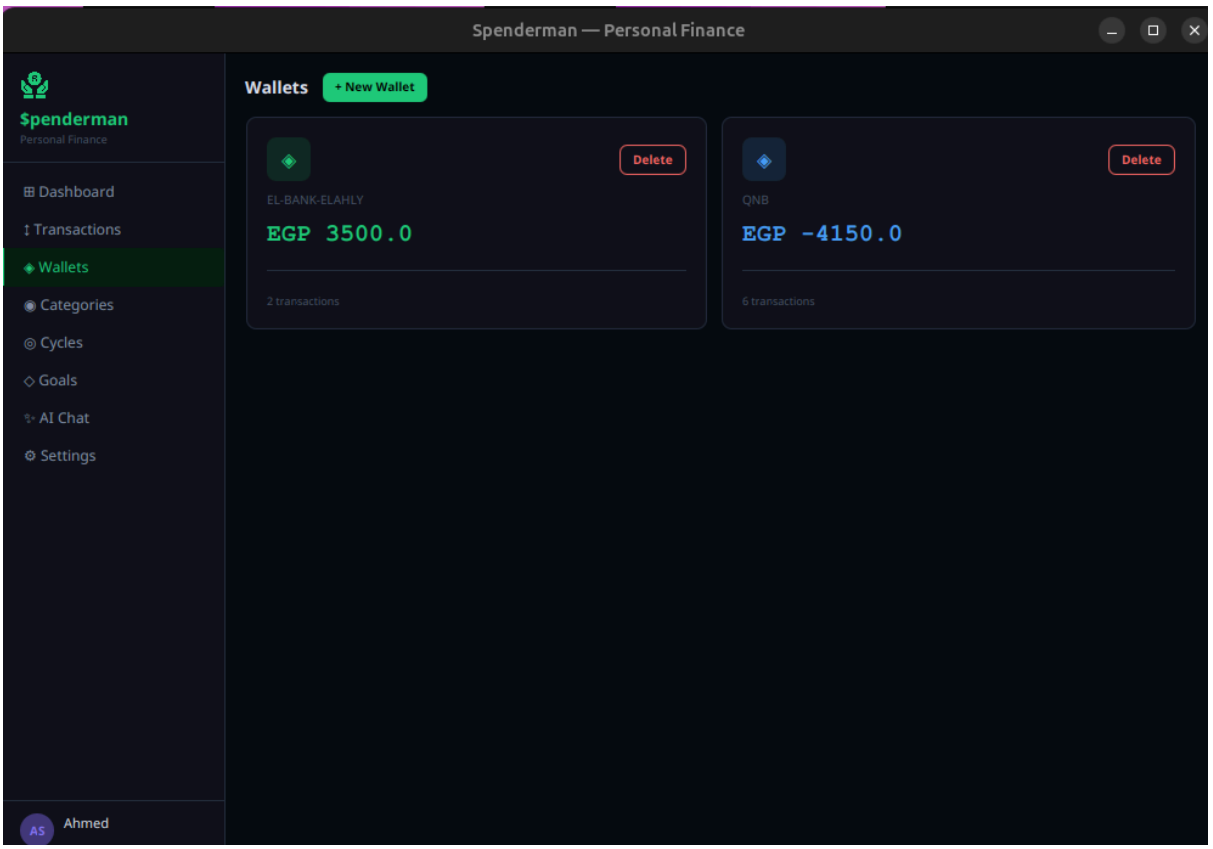
AI-Powered Financial Assistant: The application integrates an on-demand AI chatbot (ClsAiChatService) to provide users with intelligent, personalized financial advice and insights directly within the app.
Dynamic Saving Goals: Users can set specific financial milestones equipped with visual progress bars. These goals are funded exclusively through the secure transaction ledger and automatically update their status to "Completed" once the target amount is reached.
Comprehensive Settings & Theming: A dedicated settings hub empowers users to personalize their experience by toggling a persistent Dark/Light interface theme. It also handles critical security features, allowing users to safely update their passwords and manage their active login sessions.
ER-Diagram & Schema Diagram: we add the ER-Diagam of the database and the schema diagram in the SDS document.
2 additional Sequence diagrams: we make 9 sequence diagrams instead of 7.
1 additional state diagram: we make 2 state diagrams instead of 1.

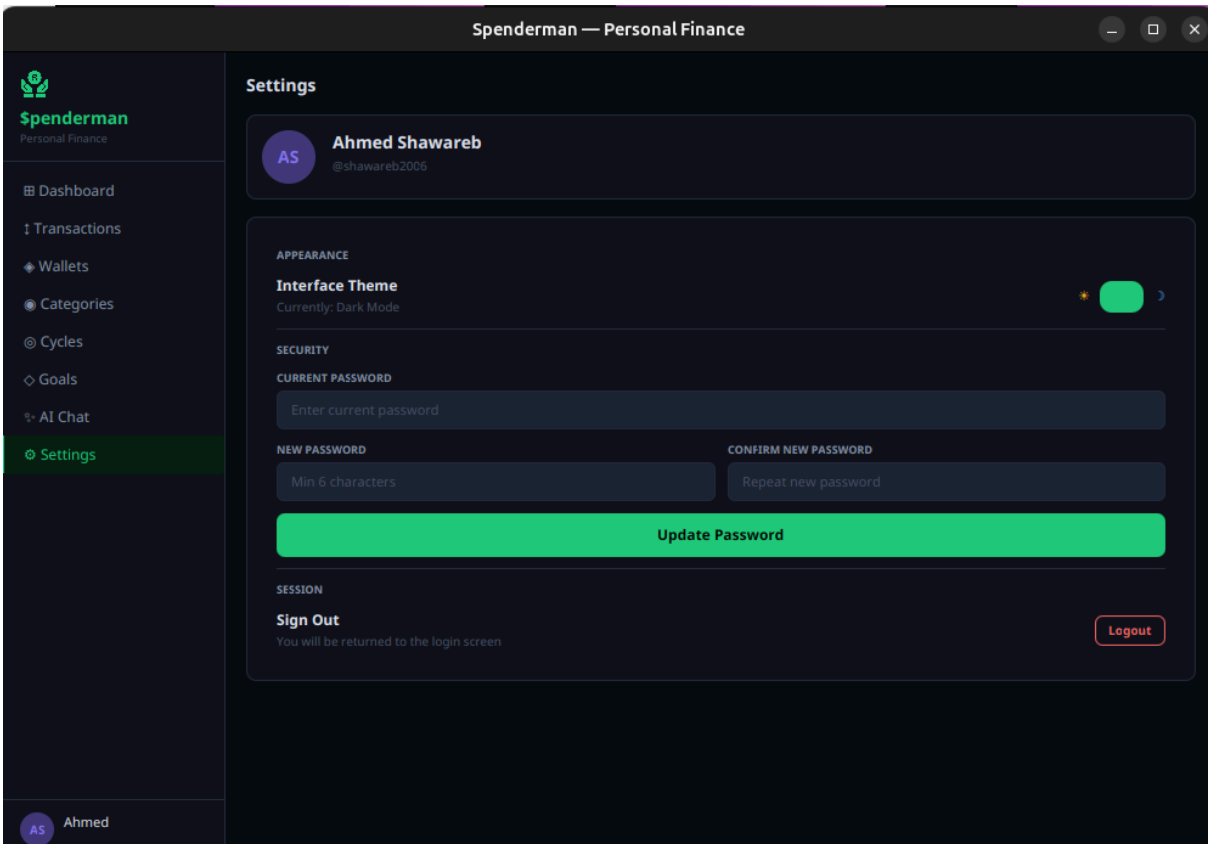
Project Snapshots






Spenderman — Personal Finance
$penderman
Personal Finance
Dashboard
Transactions
Wallets
Categories
Cycles
Goals
AI Chat
Settings
Ahmed
Transactions
+ Add
Description
Target
Category
Type
Amount
Date
Actions
Transaction
QNB
Food
Expense
-50.0 EGP
2026-05-...
Edit
Delete
Transaction
QNB
Food
Expense
-5000.0 EGP
2026-05-...
Edit
Delete
Transaction
QNB
My work
Deposit
+5000.0 EGP
2026-05-...
Edit
Delete
Transaction
QNB
Food
Expense
-5000.0 EGP
2026-05-...
Edit
Delete
Transaction
EL-BANK-ELA...
Clothes
Expense
-1500.0 EGP
2026-05-...
Edit
Delete
Transaction
QNB
My work
Deposit
+1000.0 EGP
2026-05-...
Edit
Delete
Transaction
QNB
Food
Expense
-100.0 EGP
2026-05-...
Edit
Delete
Transaction
EL-BANK-ELA...
baba
Deposit
+5000.0 EGP
2026-05-...
Edit
Delete


Spenderman — Personal Finance
$penderman
Personal Finance
Dashboard
Transactions
Wallets
Categories
Cycles
Goals
AI Chat
Settings
AS
Ahmed
Wallets
+ New Wallet
Delete
EL-BANK-ELAHLY
EGP 3500.0
2 transactions
Delete
QNB
EGP -4150.0
6 transactions


Spenderman — Personal Finance
$penderman
Personal Finance
Dashboard
Transactions
Wallets
Categories
Cycles
Goals
AI Chat
Settings
AS
Ahmed
Settings
AS
Ahmed Shawareb
@shawareb2006
APPEARANCE
Interface Theme
Currently: Dark Mode
SECURITY
CURRENT PASSWORD
Enter current password
NEW PASSWORD
Min 6 characters
CONFIRM NEW PASSWORD
Repeat new password
Update Password
SESSION
Sign Out
You will be returned to the login screen
Logout